

2

Probability

Contents

2 Probability

Introduction

2.1 Sample Spaces and Events

2.2 Axioms, Interpretations,
and Properties of Probability

2.3 Counting Techniques

2.4 Conditional Probability

2.5 Independence

2.2 Axioms, Interpretations, and Properties of Probability

Axioms, Interpretations, and Properties of Probability

Given an experiment and a sample space $\mathcal{S}$, *the objective of probability is to assign to each event A a number $P(A)$, called the probability of the event A , which will give a precise measure of the chance that A will occur.*

To ensure that the probability assignments will be consistent with our intuitive notions of probability, all assignments should satisfy the following axioms (basic properties) of probability.

Axioms, Interpretations, and Properties of Probability

Axiom 1

For any event A , $P(A) \geq 0$.

Axiom 2

$P(\mathcal{S}) = 1$.

Axiom 3

If $A_1, A_2, A_3, \dots$ is an infinite collection of disjoint events, then

$$P(A_1 \cup A_2 \cup A_3 \cup \dots) = \sum_{i=1}^{\infty} P(A_i)$$

You might wonder **why the third axiom contains no reference to a *finite* collection of disjoint events.**

Axioms, Interpretations, and Properties of Probability

It is because the corresponding property for a finite collection can be derived from our three axioms. We want our axiom list to be as short as possible and not contain any property that can be derived from others on the list.

Axiom 1 reflects the intuitive notion that the chance of A occurring should be nonnegative.

The sample space is by definition the event that must occur when the experiment is performed ($\mathcal{S}$ contains all possible outcomes), so **Axiom 2 says that the maximum possible probability of 1 is assigned to $\mathcal{S}$.**

Axioms, Interpretations, and Properties of Probability

The third axiom formalizes the idea that if we wish the probability that at least one of a number of events will occur and no two of the events can occur simultaneously, then the chance of at least one occurring is the sum of the chances of the individual events.

Proposition

$P(\emptyset) = 0$ where $\emptyset$ is the null event (the event containing no outcomes whatsoever).

This in turn implies that the property contained in Axiom 3 is valid for a *finite* collection of disjoint events.

Axioms, Interpretations, and Properties of Probability

Proof First consider the infinite collection $A_1 = \emptyset, A_2 = \emptyset, A_3 = \emptyset, \dots$. Since $\emptyset \cap \emptyset = \emptyset$, the events in this collection are disjoint and $\bigcup A_i = \emptyset$. The third axiom then gives

$$P(\emptyset) = \sum P(\emptyset)$$

This can happen only if $P(\emptyset) = 0$.

Now suppose that $A_1, A_2, \dots, A_k$ are disjoint events, and append to these the infinite collection $A_{k+1} = \emptyset, A_{k+2} = \emptyset, A_{k+3} = \emptyset, \dots$. Again invoking the third axiom,

$$P\left(\bigcup_{i=1}^k A_i\right) = P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) = \sum_{i=1}^k P(A_i)$$

Example

Consider tossing a thumbtack in the air. When it comes to rest on the ground, either its point will be up (the outcome U) or down (the outcome D). The sample space for this event is therefore $\mathcal{S} = \{U, D\}$.

The axioms specify $P(\mathcal{S}) = 1$, so the probability assignment will be completed by determining $P(U)$ and $P(D)$.

Since U and D are disjoint and their union is $\mathcal{S}$, the foregoing proposition implies that

$$1 = P(\mathcal{S}) = P(U) + P(D)$$

Example

cont'd

It follows that $P(D) = 1 - P(U)$.

One possible assignment of probabilities is

$$P(U) = .5, P(D) = .5,$$

whereas another possible assignment is

$$P(U) = .75, P(D) = .25.$$

In fact, letting p represent any fixed number between 0 and 1, $P(U) = p, P(D) = 1 - p$ is an assignment consistent with the axioms.

Example

Consider testing batteries coming off an assembly line one by one until one having a voltage within prescribed limits is found.

The simple events are

$$E_1 = \{S\},$$

$$E_2 = \{FS\},$$

$$E_3 = \{FFS\},$$

$$E_4 = \{FFFS\}, \dots$$

Suppose the probability of any particular battery being satisfactory is .99.

Example

cont'd

Then it can be shown that

$$P(E_1) = .99, \quad P(E_2) = (.01)(.99),$$

$P(E_3) = (.01)^2(.99), \dots$ is an assignment of probabilities to the simple events that satisfies the axioms. In particular, because the E_i s are disjoint and $\mathcal{S} = E_1 \cup E_2 \cup E_3 \cup \dots$, it must be the case that

$$\begin{aligned} 1 &= P(S) = P(E_1) + P(E_2) + P(E_3) + \dots \\ &= .99[1 + .01 + (.01)^2 + (.01)^3 + \dots] \end{aligned}$$

Example

cont'd

Here we have used the formula for the sum of a geometric series:

$$a + ar + ar^2 + ar^3 + \dots = \frac{a}{1 - r}$$

However, another legitimate (according to the axioms) probability assignment of the same “geometric” type is obtained by replacing .99 by any other number p between 0 and 1 (and .01 by $1 - p$).



Interpreting Probability

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Interpreting Probability

Previous examples show that the axioms do not completely determine an assignment of probabilities to events. The axioms serve only to rule out assignments inconsistent with our intuitive notions of probability.

In the tack-tossing experiment, two particular assignments were suggested.

The appropriate or correct assignment depends on the nature of the thumbtack and also on one's interpretation of probability.

Interpreting Probability

The interpretation that is most frequently used and most easily understood is based on the **notion of relative frequencies**.

Consider an experiment that can be repeatedly performed in an identical and independent fashion, and let A be an event consisting of a fixed set of outcomes of the experiment.

Simple examples of such repeatable experiments include the tack-tossing and die-tossing experiments previously discussed.

Interpreting Probability

If the experiment is performed n times, on some of the replications the event A will occur (the outcome will be in the set A), and on others, A will not occur.

Let $n(A)$ denote the number of replications on which A does occur.

Then the ratio $n(A)/n$ is called the *relative frequency* of occurrence of the event A in the sequence of n replications.

Interpreting Probability

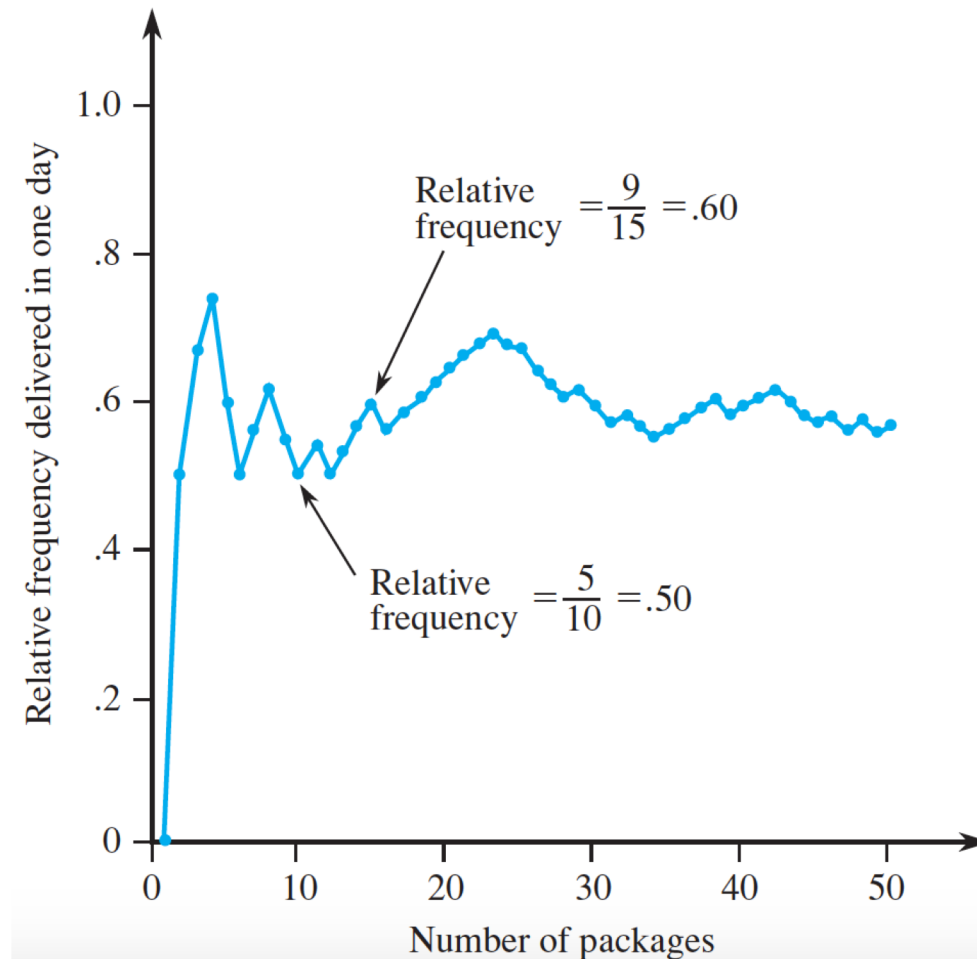
For example, let A be the event that a package sent within the state of California for 2nd day delivery actually arrives within one day.

The results from sending 10 such packages (the first 10 replications) are as follows:

Package #	1	2	3	4	5	6	7	8	9	10
Did A occur?	N	Y	Y	Y	N	N	Y	Y	N	N
Relative frequency of A	0	.5	.667	.75	.6	.5	.571	.625	.556	.5

Interpreting Probability

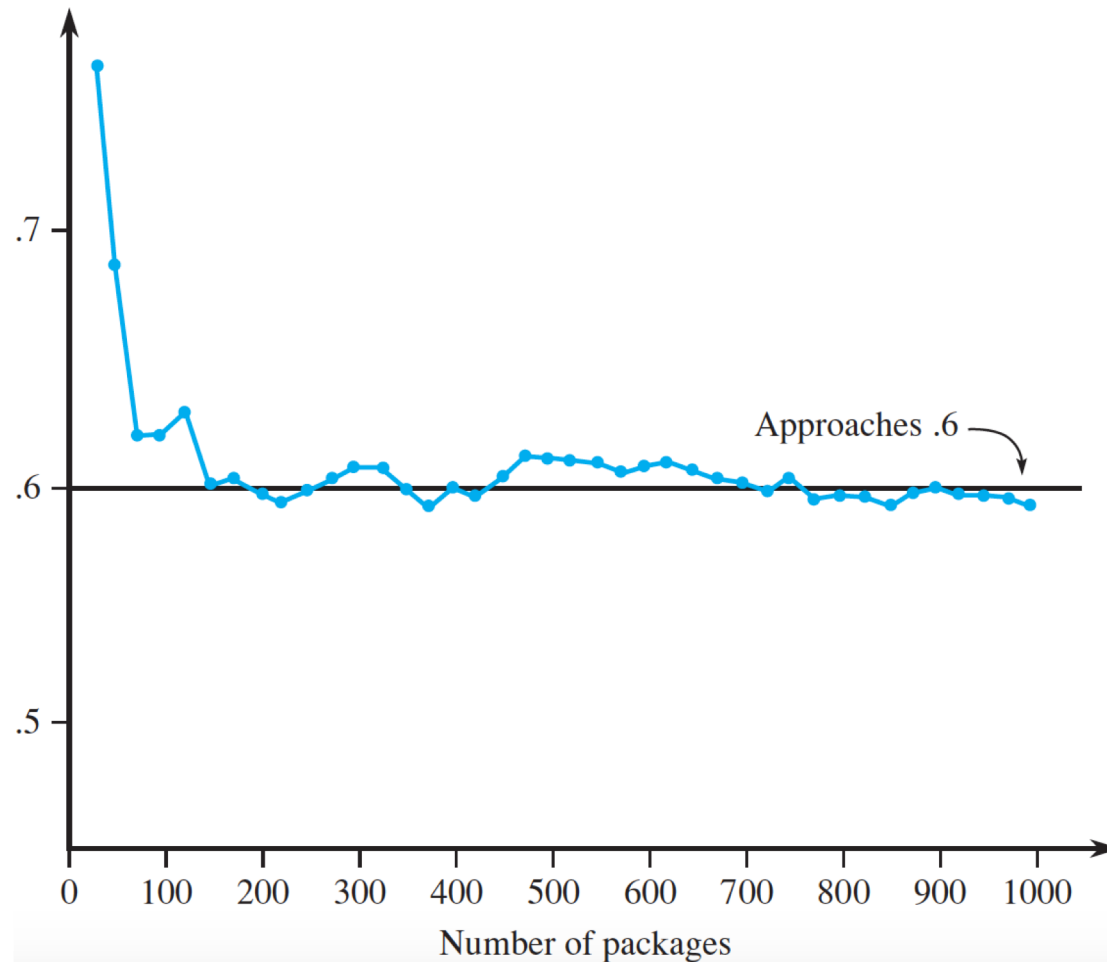
How the relative frequency $n(A)/n$ fluctuates rather substantially over the course of the first 50 replications.



Behavior of relative frequency (a) Initial fluctuation

Interpreting Probability

But as the number of replications continues to increase, The relative frequency stabilizes as shown in the figure.



Behavior of relative frequency (b) Long-run stabilization

Interpreting Probability

More generally, empirical evidence, based on the results of many such **repeatable experiments**, indicates that **any relative frequency of this sort will stabilize as the number of replications n increases.**

That is, as n gets arbitrarily large, $n(A)/n$ approaches a limiting value referred to as the ***limiting (or long-run) relative frequency of the event A .***

The ***objective interpretation of probability*** identifies this limiting relative frequency with $P(A)$.

Interpreting Probability

Suppose that probabilities are assigned to events in accordance with their limiting relative frequencies.

Then a statement such as “the probability of a package being delivered within one day of mailing is .6” means that of a large number of mailed packages, roughly 60% will arrive within one day.

Similarly, if B is the event that an appliance of a particular type will need service while under warranty, then $P(B) = .1$ is interpreted to mean that in the long run 10% of such appliances will need warranty service.

Interpreting Probability

This doesn't mean that exactly 1 out of 10 will need service, or that exactly 10 out of 100 will need service, because 10 and 100 are not the long run.

This relative frequency interpretation of probability is said to be **objective** because it rests on a property of **the experiment rather than on any particular individual** concerned with the experiment.

For example, two different observers of a sequence of coin tosses should both use the same probability assignments since the observers have nothing to do with limiting relative frequency.

Interpreting Probability

In practice, this interpretation is not as objective as it might seem, since the limiting relative frequency of an event will not be known.

Thus we will have to assign probabilities based on our beliefs about the limiting relative frequency of events under study.

Fortunately, **there are many experiments for which there will be a consensus** with respect to probability assignments.

Interpreting Probability

When we speak of a fair coin, we shall mean

$$P(H) = P(T) = .5,$$

and a fair die is one for which limiting relative frequencies of the six outcomes are all $\frac{1}{6}$, suggesting probability assignments $P(\{1\}) = \cdots = P(\{6\}) = \frac{1}{6}$.

Because the **objective interpretation** of probability is based on the notion of limiting frequency, **its applicability is limited to experimental situations that are repeatable.**

Interpreting Probability

Yet the language of probability is often used in connection with situations that are inherently unrepeatable.

Examples include: “The chances are good for a peace agreement”; “It is likely that our company will be awarded the contract”; and “Because their best quarterback is injured, I expect them to score no more than 10 points against us.”

In such situations we would like, as before, to assign numerical probabilities to various outcomes and events (e.g., the probability is .9 that we will get the contract).

Interpreting Probability

We must therefore adopt an alternative interpretation of these probabilities. **Because different observers may have different prior information and opinions** concerning such experimental situations, probability assignments may now **differ from individual to individual.**

Interpretations in such situations are thus referred to as ***subjective***.

The book by Robert Winkler listed in the chapter references gives a very readable survey of several subjective interpretations.



More Probability Properties

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More Probability Properties

Proposition

For any event A , $P(A) + P(A') = 1$, from which $P(A) = 1 - P(A')$.

Proof In Axiom 3, let $k = 2$, $A_1 = A$, and $A_2 = A'$. Since by definition of A' , $A \cup A' = \mathcal{S}$ while A and A' are disjoint, $1 = P(\mathcal{S}) = P(A \cup A') = P(A) + P(A')$. ■

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Example

Consider a system of five identical components connected in series, as illustrated in Figure 2.3.

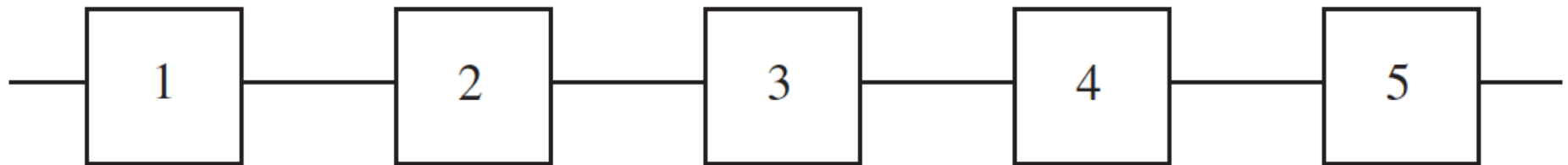


Figure 2.3 A system of five components connected in a series

Denote a component that fails by F and one that doesn't fail by S (for success).

Let A be the event that the system fails.

Calculate $P(A)$.

For A to occur, at least one of the individual components must fail.

Example

cont'd

Outcomes in A include $SSFSS$ (1, 2, 4, and 5 all work, but 3 does not), $FFSSS$, and so on.

Two assumptions:

We will see in Section 2.5 that if 90% of all such components do not fail ($P(S) = 0.9$) and **different components fail independently** (**Assumption**) of one another, then

$$P(A') = P(SSSSS) = .9^5 = .59.$$

Thus $P(A) = 1 - .59 = .41$; so among a large number of such systems, roughly 41% will fail.

More Probability Properties

In general, the foregoing proposition is useful when the event of interest can be expressed as “at least . . . ,” since then the complement “less than . . .” may be easier to work with (in some problems, “more than . . .” is easier to deal with than “at most . . .”).

When you are having difficulty calculating $P(A)$ directly, think of determining $P(A')$.

PROPOSITION

For any event A , $P(A) \leq 1$.

This is because $1 = P(A) + P(A') \geq P(A)$ since $P(A') \geq 0$.

More Probability Properties

When events A and B are **mutually exclusive**,

$$P(A \cup B) = P(A) + P(B).$$

For events that are not mutually exclusive, adding $P(A)$ and $P(B)$ results in “double-counting” outcomes in the intersection. The next result shows how to correct for this.

Proposition

For any two events A and B ,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

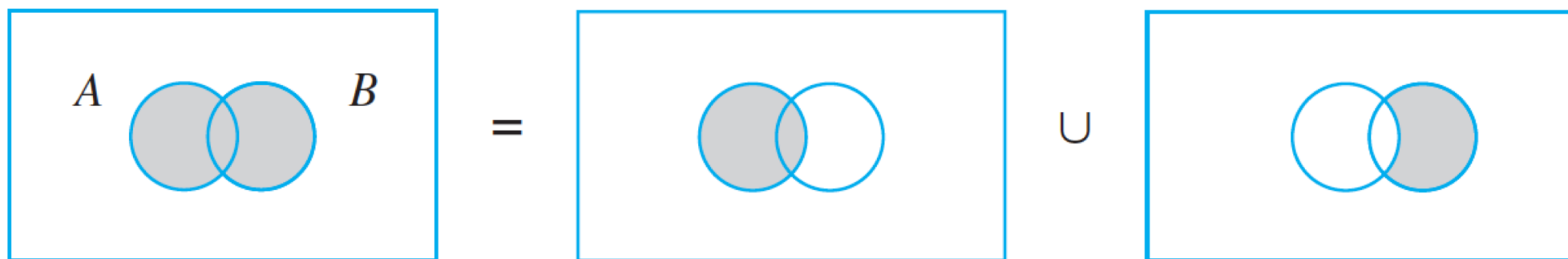
More Probability Properties

PROPOSITION

For any two events A and B ,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\begin{aligned} P(A \cup B) &= P(A) + P(B \cap A') = P(A) + [P(B) - P(A \cap B)] \\ &= P(A) + P(B) - P(A \cap B) \end{aligned}$$



More Probability Properties

PROPOSITION

For any three events A , B , and C ,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

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Determining Probabilities Systematically

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Determining Probabilities Systematically

Consider a sample space that is either finite or “countably infinite” (the latter means that outcomes can be listed in an infinite sequence, so there is a first outcome, a second outcome, a third outcome, and so on—for example, the battery testing scenario of Example 12).

Let $E_1, E_2, E_3, \dots$ denote the corresponding simple events, each consisting of a single outcome.

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Determining Probabilities Systematically

A sensible strategy for probability computation is to first determine each **simple event** probability, with the requirement that $\sum P(E_i) = 1$.

$$\sum P(E_i) = 1$$

Then the probability of any **compound event** A is computed by adding together the $P(E_i)$'s for all E_i 's in A :

$$P(A) = \sum_{\text{all } E_i \text{'s in } A} P(E_i)$$

Example

During off-peak hours a commuter train has five cars. Suppose a commuter is twice as likely to select the middle car (#3) as to select either adjacent car (#2 or #4), and is twice as likely to select either adjacent car as to select either end car (#1 or #5).

The probability that one of the three middle cars is selected ?

Let $p_i = P(\text{car } i \text{ is selected}) = P(E_i)$. Then we have $p_3 = 2p_2 = 2p_4$ and $p_2 = 2p_1 = 2p_5 = p_4$. This gives

$$1 = \sum P(E_i) = p_1 + 2p_1 + 4p_1 + 2p_1 + p_1 = 10p_1$$

implying $p_1 = p_5 = .1$, $p_2 = p_4 = .2$, $p_3 = .4$. The probability that one of the three middle cars is selected (a compound event) is then $p_2 + p_3 + p_4 = .8$.



Equally Likely Outcomes

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Equally Likely Outcomes

In many experiments consisting of N outcomes, it is reasonable to assign equal probabilities to all N simple events.

These include such obvious examples as tossing a fair coin or fair die once or twice (or any fixed number of times), or selecting one or several cards from a well-shuffled deck of 52. With $p = P(E_i)$ for every i ,

$$1 = \sum_{i=1}^N P(E_i) = \sum_{i=1}^N p = p \cdot N \quad \text{so } p = \frac{1}{N}$$

That is, if there are N equally likely outcomes, the probability for each is $1/N$.

Equally Likely Outcomes

Now consider an event A , with $N(A)$ denoting the number of outcomes contained in A . Then

$$P(A) = \sum_{E_i \text{ in } A} P(E_i) = \sum_{E_i \text{ in } A} \frac{1}{N} = \frac{N(A)}{N}$$

Thus **when outcomes are equally likely**, computing probabilities reduces to counting: determine both the number of outcomes $N(A)$ in A and the number of outcomes N in $\mathcal{S}$, and form their ratio.

Example

You have six unread mysteries on your bookshelf and six unread science fiction books.

The first three of each type are hardcover, and the last three are paperback.

Consider randomly selecting one of the six mysteries and then randomly selecting one of the six science fiction books to take on a post-finals vacation to Acapulco (after all, you need something to read on the beach).

Number the mysteries 1, 2, . . . , 6, and do the same for the science fiction books.

Example

cont'd

Then each outcome is a pair of numbers such as (4, 1), and there are $N = 36$ possible outcomes (For a visual of this situation, refer the table below and delete the first row and column).

		<i>Second Station</i>						
		0	1	2	3	4	5	6
<i>First Station</i>	0	(0, 0)	(0, 1)	(0, 2)	(0, 3)	(0, 4)	(0, 5)	(0, 6)
	1	(1, 0)	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
	2	(2, 0)	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
	3	(3, 0)	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
	4	(4, 0)	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
	5	(5, 0)	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
	6	(6, 0)	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

Example

cont'd

With random selection as described, the 36 outcomes are equally likely.

Nine of these outcomes are such that both selected books are paperbacks (those in the lower right-hand corner of the referenced table): (4, 4), (4, 5), . . . , (6, 6).

So the probability of the event A that both selected books are paperbacks is

$$P(A) = \frac{N(A)}{N} = \frac{9}{36} = .25$$